

Test Paper on OOPS concept

1. **What is Object-Oriented Programming? List and briefly explain its four main principles.**
2. **Differentiate between abstraction and encapsulation with real-world examples.**
3. **What is inheritance in Java? What are the types of inheritance supported?**
4. **What is method overloading? Write a simple Java code to demonstrate it.**
5. **What is the role of the `super` keyword in Java? Give an example.**
6. **Write a Java program to create a class `Person` with attributes `name` and `age`. Use a constructor to initialize values and a method to display them. Create an object in the `main` method.

Create a base class `Vehicle` with a method `start()`. Inherit it into two classes: `Car` and `Bike`, both overriding the `start()` method. Demonstrate runtime polymorphism.**
7. **Design a class `BankAccount` with private data members `accountNumber` and `balance`. Use getters and setters to access and update them. Show how encapsulation is achieved.**
9. **Write a Java program to demonstrate the concept of abstraction:**

Create an abstract class `Shape` with an abstract method `calculateArea()`.

Create two subclasses `Circle` and `Rectangle` that implement the method.

Include constructors and methods to input and display area.

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10.Design a class hierarchy to represent employees:

Base class: Employee (fields: id, name)

Subclass: Manager (additional field: department)

Subclass: Developer (additional field: language)

**Each class should have constructors and a method
displayInfo().**

Create objects of each class and demonstrate method overriding.